

MA341

APPLIED ODE'S

Handout 3

Definition 1. A Separable equation is an equation that can be transformed into $\frac{dy}{dx} + f(x)g(y) = 0$. One rewrite it as $\frac{dy}{g(y)} = -f(x)dx$ to solve by integration. A **singular solutions** y_0 may happen if $g(y_0) = 0$.

Definition 2. (Linear Substitution) Suppose $y' = f(ax + by + c)$ define $z = ax + by + c$ to get a separable equation $z' = a + by' = a + f(z)$.

Definition 3. (Homogeneous equations) A function $f(x, y)$ is called homogeneous if $f(tx, ty) = f(x, y)$. Particularly, in such case $f(x, y) = f(\frac{x}{y})$. To solve ODE of the form $y' = f(x, y) = f(\frac{x}{y})$ substitute $z(x) = \frac{y(x)}{x}$, now $y = xz$ and $y' = xz' + z = f(z)$, and we got a separable equation $\frac{dz}{f(z)-z} = \frac{dx}{x}$.

Definition 4. (Quotient of two linear functions) Suppose $y' = f\left(\frac{ax+by+c}{\alpha x+\beta y+\gamma}\right)$ If the lines are parallel $\alpha = ta, \beta = tb$, substitute $v(x) = ax + by$, to solve $v' = a + by' = a + bf\left(\frac{v+c}{tv+\gamma}\right)$. If the lines intersects at point (x_0, y_0) , define $t = x - x_0$ and $s = y - y_0$, to get an homogeneous equation

$$\frac{ds}{dt} = \frac{dy}{dx} = f\left(\frac{ax+by+c}{\alpha x+\beta y+\gamma}\right) = f\left(\frac{at+bs}{\alpha t+\beta s}\right) = f\left(\frac{a+b\frac{s}{t}}{\alpha+\beta\frac{s}{t}}\right)$$

Next, a substitution $s(t) = tz(t)$ leads to $\frac{ds}{dt} = tz' + z = f\left(\frac{a+bz}{\alpha+\beta z}\right)$ so we solve $\left(f\left(\frac{a+bz}{\alpha+\beta z}\right) - z\right)^{-1} dz = \frac{dt}{t}$.

Definition 5. Exact equation is an equation that can be transformed into $P(x, y)dx + Q(x, y)dy = 0$, where $P_y = Q_x$. To find it, solve $f(x, y) = \int Pdx = \tilde{P} + g(y)$ to determine g , solve $f_y = Q$. **Note:** f is a potential of a conservative vector field: $\langle P, Q \rangle = \nabla f$.

Definition 6. Integration factor Given a non exact equation $Mdx + Ndy = 0$ sometimes it is possible to find an integration factor μ such that $\mu(Mdx + Ndy) = (\mu M)dx + (\mu N)dy = 0$ is exact. Here is the possible cases:

$$\begin{aligned}\frac{\mu'}{\mu} &= \frac{M_y - N_x}{N} = f(x) \Rightarrow \mu(x) = e^{\int_0^x f(s)ds} \\ \frac{\mu'}{\mu} &= \frac{N_x - M_y}{M} = g(y) \Rightarrow \mu(x) = e^{\int_0^y g(s)ds} \\ \frac{\mu'}{\mu} &= \frac{N_x - M_y}{xM - yN} = h(xy) \Rightarrow \mu(x) = e^{\int_0^x h(s)ds}\end{aligned}$$